

Computer & ICT Fundamentals Notes

Comprehensive Revision & Study Guide | Information Technology, Operating Systems & Self-Management

1. Introduction to IT & ITES

Information Technology (IT): Plays a significant role in modern day-to-day activities. It assists in easy analysis of data/information for decision making, creating efficient processes, automating business operations, and reducing operational waste.

IT Enabled Services (ITES): Web-enabled or remote services that leverage information technology to improve the overall efficiency of an organization. ITES provides career opportunities across software development, MIS (Management Information System), call centers, customer support, medical transcription, and healthcare support.

Structure of the IT-BPM Industry

The IT Business Process Management (IT-BPM) industry comprises companies categorized as follows:

- **Multinational Companies (MNCs):** Headquarters outside India but operating globally (including India), catering to both international and domestic clients.
- **Indian Service Providers (ISPs):** Started and headquartered in India with multiple international client operations and offices.
- **Global In-house Centers (GICs):** Operations perform solely for the parent company and do not cater to external clients, allowing cost-effective internal operations.

Difference between BPO and BPM

Feature	BPO (Business Process Outsourcing)	BPM (Business Process Management)
Definition	Transferring specific business processes or operational activities to a third-party service provider.	A systematic management approach to analyze, design, maintain, and optimize business processes.
Primary Goal	Focuses on reducing operational costs by executing specialized tasks externally.	Focuses on optimizing internal workflows to make them faster, efficient, and beneficial.
Scope	Limited to specific outsourced operational functions.	Covers the complete flow of end-to-end business activities.
Management	Managed by external third-party vendor management teams.	Managed internally by the company's core management team.

2. ICT Applications & Peripherals

ICT (Information and Communication Technology): A combination of communication and IT used to create, store, manipulate, transmit, and process information.

- **Administration:** Digital identity cards, e-Challan systems, online schemes, and income tax filing.

- **Healthcare:** E-Doctors, online medicine ordering, fitness tracking via wearables, digital vital measurement, e-surgery & robotic surgery.
- **Education & Daily Life:** E-learning portals, digital notes, online classes, AI learning tools, student attendance systems.
- **E-Banking:** Online fund transfers, account tracking, bill payments, digital receipts, and self-service banking options.

Computer Operations & Port Standards

Basic Operation	Description	Associated Components / Ports
Input	Entering raw data and instructions into the computer.	Keyboard, Mouse, Scanner, Web Cam, Digitizer.
Processing	Executing mathematical & logical operations via Central Processing Unit (CPU).	CU: Control Unit, ALU: Arithmetic Logic Unit, MU: Memory Unit.
Output & Storage	Presenting processed data and persisting it into primary/secondary memory.	Monitor, Printer, Hard Drives, Flash Storage.
Port Types	Physical interfaces connecting external peripherals.	Serial Port (9/25 pin), Parallel Port (25 pin), USB (Universal Serial Bus), Ethernet (RJ45), HDMI (High Definition Multimedia Interface).

3. Operating Systems & Software

An **Operating System (OS)** is primary software that acts as an interface between the user and computer hardware. Without an OS, computer hardware cannot execute application programs.

Core OS Functions

- **Process Management:** Controls, schedules, and allocates resources to active processes and tasks.
- **Memory Management:** Manages primary (RAM) and secondary storage allocations for efficient process execution.
- **Device Management:** Controls I/O devices and storage hardware using device drivers.
- **File Management:** Organizes files and folders, providing structured file access and security.
- **Bootting:** The sequence of loading the operating system into primary memory upon switching on the computer.

CUI vs GUI Operating Systems

Feature	Console User Interface (CUI)	Graphical User Interface (GUI)
Interface Type	Primarily text-based command prompt interface.	Visual interface containing icons, windows, menus, and text.

Feature	Console User Interface (CUI)	Graphical User Interface (GUI)
Processing	Processed entirely by the central processor (CPU).	Processed with support from Graphics Processing Units (GPU).
User Interaction	Requires memorizing specific keyboard commands.	Mouse pointer & intuitive visual touch/click controls.
Examples	MS-DOS, Unix Command Shell.	Windows 10/11, Android, macOS, Linux (Ubuntu Desktop).

4. Self-Management & Personal Skills

Self-Management: The ability to regulate one's emotions, behaviors, and physical/mental states to take charge of personal development, maintain self-control, and achieve life goals.

Key Self-Management Skills & Qualities

- **Self-Confidence:** Belief in one's capability to handle tasks and handle challenging situations.
- **Positive Thinking:** A constructive, optimistic mental attitude focusing on solutions rather than dwelling on problems.
- **Goal Setting:** Defining targets within specific deadlines to complete tasks efficiently without procrastination.
- **Strength & Weakness Analysis:** Identifying natural abilities, areas of interest, and identifying areas needing improvement.
- **Time Management:** Planning and prioritizing daily routines to maximize productivity.

Factors Boosting Self-Confidence	Factors Decreasing Self-Confidence
Looking at positive outcomes and learning from mistakes.	Thinking "I cannot do this task" before attempting it.
Surrounding oneself with supportive, optimistic people.	Fixating over past failures instead of treating them as lessons.
Hard work, commitment, and dedicated practice.	Lack of goal setting, poor personal hygiene, and negative environment.